

MVLU COLLEGE
A.I. Practical No.9

Aim:- Association Rule Mining

Implement the Association Rule Mining algorithm (e.g., Apriori) to find frequent itemsets.

Generate association rules from the frequent itemsets

Code:-

```
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import pandas as pd
import matplotlib.pyplot as plt
from mixtend.preprocessing import TransactionEncoder
from mixtend.frequent_patterns import apriori, association_rules

file_path = r"F:\Downloads\Most Runs - 2010.csv"
df = pd.read_csv(file_path)
print("Hariprasad Vishwakarma T127")
print("Dataset loaded successfully!")
print("\nFirst 5 rows:")
print(df.head())
print("\nDataset shape:")
print(df.shape)
print("\nColumn names:")
print(df.columns.tolist())

if "Player" in df.columns:
    df = df.drop("Player", axis=1)

if "POS" in df.columns:
    df = df.drop("POS", axis=1)

if "HS" in df.columns:
    df["HS"] = df["HS"].astype(str).str.replace(" ", "", regex=False)
    df["HS"] = pd.to_numeric(df["HS"], errors="coerce")

for column in df.columns:
    df[column] = pd.to_numeric(df[column], errors="coerce")

df = df.dropna()

print("\nMissing values removed")

transactions = []

for _, row in df.iterrows():
    transaction = []

    if row["Runs"] >= 400:
        transaction.append("High Runs")
    elif row["Runs"] >= 250:
        transaction.append("Medium Runs")
```

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if row["Runs"] >= 400:
    transaction.append("High Runs")
elif row["Runs"] >= 250:
    transaction.append("Medium Runs")
else:
    transaction.append("Low Runs")

if row["Avg"] >= 40:
    transaction.append("High Average")
elif row["Avg"] >= 25:
    transaction.append("Medium Average")
else:
    transaction.append("Low Average")

if row["SR"] >= 140:
    transaction.append("High Strike Rate")
elif row["SR"] >= 120:
    transaction.append("Medium Strike Rate")
else:
    transaction.append("Low Strike Rate")

if row["100"] > 0:
    transaction.append("Century")
if row["50"] > 0:
    transaction.append("Fifty")
if row["4s"] >= 30:
    transaction.append("Many Fours")
if row["6s"] >= 15:
    transaction.append("Many Sixes")

transactions.append(transaction)

print("\nNumber of transactions:", len(transactions))

te = TransactionEncoder()
te_array = te.fit(transactions).transform(transactions)

basket = pd.DataFrame(
    te_array,
    columns=te.columns_
)

transactions.append(transaction)

print("\nNumber of transactions:", len(transactions))

te = TransactionEncoder()
te_array = te.fit(transactions).transform(transactions)

basket = pd.DataFrame(
    te_array,
    columns=te.columns_
)

print("\nTransaction Matrix:")
print(basket.head())

frequent_itemsets = apriori(
    basket,
    min_support=0.10,
    use_colnames=True
)

frequent_itemsets["itemsets_length"] = frequent_itemsets["itemsets"].apply(len)

print("\nFrequent Itemsets:")
print(
    frequent_itemsets.sort_values(
        by="support",
        ascending=False
    ).head(20)
)

rules = association_rules(
    frequent_itemsets,
    metric="Confidence",
    min_threshold=0.30
)

print("\nAssociation Rules:")
print(rules[
    "antecedents",
    "consequents",
    "support",
    "confidence",
    "lift",
    "leverage",
    "conviction",
    "all_confidence_lift",
    "all_leverage",
    "all_conviction"
])
```

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"consequents",
"support",
"confidence",
"lift"
]).head(20).to_string(index=False))

rules = rules.sort_values(
    by="lift",
    ascending=False
)

strong_rules = rules[
    (rules["confidence"] >= 0.50) &
    (rules["lift"] > 1)
]

print("\nStrong Association Rules:")
print(
    strong_rules[["antecedents",
"consequents",
"support",
"confidence",
"lift"
]].head(20).to_string(index=False)
)

top_itemsets = frequent_itemsets.sort_values(
    by="support",
    ascending=False
).head(10)

itemset_labels = [
    ", ".join(list(itemset))
    for itemset in top_itemsets["itemsets"]
]

plt.figure(figsize=(10, 6))

plt.barh(
    itemset_labels[::-1],
    top_itemsets["support"].values[::-1]
)
```

```
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top_itemsets["support"].values[::-1]
)

plt.xlabel("Support")
plt.ylabel("Itemset")
plt.title("Top 10 Frequent Itemsets")

plt.tight_layout()

top_rules = rules.head(10).copy()

rule_labels = [
    ", ".join(list(row["antecedents"])) +
    " -> " +
    ", ".join(list(row["consequents"]))
    for _, row in top_rules.iterrows()
]

plt.figure(figsize=(10, 6))

plt.barh(
    rule_labels[::-1],
    top_rules["lift"].values[::-1]
)

plt.xlabel("Lift")
plt.ylabel("Association Rule")
plt.title("Top 10 Association Rules by Lift")

plt.tight_layout()

print("\n=====")
print("ASSOCIATION RULE MINING SUMMARY")
print("=====")
print("Total transactions:", len(transactions))
print("Number of frequent itemsets:", len(frequent_itemsets))
print("Number of association rules:", len(rules))
print("Number of strong rules:", len(strong_rules))
print("=====")

plt.show()
```

Output:-

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```
"IDLE Shell 3.13.5"
File Edit Shell Debug Options Window Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: D:/Hari/A.I/Practical 2.py =====
Hariprasad Vishwakarma T127
=== A* Search Algorithm ===
Optimal Path Found:
Lucknow -> Ayodhya -> Azamgarh -> Varanasi
Total Distance = 40 km

===== RESTART: D:/Hari/A.I/Practical 9.py =====
Hariprasad Vishwakarma T127
Dataset loaded successfully!

First 5 rows:
  POS  Player  Mat  Inns  NO  Runs  ...  BF  SR  100  50  4s  6s
0    1  Sachin Tendulkar  15   15  2   618  ...  466  132.61  0  5  86  3
1    2   Jacques Kallis  16   16  4   572  ...  494  115.78  0  6  67  9
2    3    Suresh Raina  16   16  5   520  ...  364  142.85  0  4  45  22
3    4   Sourav Ganguly  14   14  1   493  ...  419  117.66  0  4  58  15
4    5    Murali Vijay  15   15  2   458  ...  292  156.84  1  2  36  26

[5 rows x 14 columns]

Dataset shape:
(150, 14)

Column names:
['POS', 'Player', 'Mat', 'Inns', 'NO', 'Runs', 'HS', 'Avg', 'BF', 'SR', '100', '50', '4s', '6s']

Missing values removed

Number of transactions: 150

Transaction Matrix:
  Century  Fifty  ...  Medium Runs  Medium Strike Rate
0    False   True  ...         False                True
1    False   True  ...         False                False
2    False   True  ...         False                False
3    False   True  ...         False                False
4     True   True  ...         False                False

[5 rows x 13 columns]

Frequent Itemsets:
  support  itemsets  itemsets_length
3  0.793333  (Low Runs)  1
2  0.686667  (Low Average)  1
15 0.653333  (Low Runs, Low Average)  2
4  0.600000  (Low Strike Rate)  1
18 0.560000  (Low Runs, Low Strike Rate)  2
16 0.506667  (Low Strike Rate, Low Average)  2
23 0.500000  (Low Strike Rate, Low Runs, Low Average)  3
0  0.300000  (Fifty)  1
6  0.246667  (Medium Average)  1
8  0.226667  (Medium Strike Rate)  1
12 0.186667  (Fifty, Medium Average)  2
1  0.173333  (High Strike Rate)  1
20 0.153333  (Low Runs, Medium Strike Rate)  2
7  0.146667  (Medium Runs)  1
13 0.133333  (Fifty, Medium Runs)  2
5  0.126667  (Many Fours)  1
11 0.126667  (Fifty, Many Fours)  2
21 0.113333  (Medium Average, Medium Runs)  2
14 0.113333  (Fifty, Medium Strike Rate)  2
22 0.113333  (Fifty, Medium Average, Medium Runs)  3

Association Rules:
  antecedents  consequents  support  confidence  lift
(Fifty)  (High Strike Rate)  0.106667  0.355556  2.051282
(High Strike Rate)  (Fifty)  0.106667  0.615385  2.051282
(Fifty)  (Low Runs)  0.106667  0.355556  0.448179
(Fifty)  (Many Fours)  0.126667  0.422222  3.333333
```

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```
"IDLE Shell 3.13.5"
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Association Rules:
antecedents      consequents support confidence lift
(Fifty) (High Strike Rate) 0.106667 0.355556 2.051282
(High Strike Rate) (Fifty) 0.106667 0.615385 2.051282
(Fifty) (Low Runs) 0.106667 0.355556 0.448179
(Fifty) (Many Fours) 0.126667 0.422222 3.333333
(Many Fours) (Fifty) 0.126667 1.000000 3.333333
(Fifty) (Medium Average) 0.186667 0.622222 2.522523
(Medium Average) (Fifty) 0.186667 0.756757 2.522523
(Fifty) (Medium Runs) 0.133333 0.444444 3.030303
(Medium Runs) (Fifty) 0.133333 0.909091 3.030303
(Fifty) (Medium Strike Rate) 0.113333 0.377778 1.666667
(Medium Strike Rate) (Fifty) 0.113333 0.500000 1.666667
(Low Runs) (Low Average) 0.653333 0.823529 1.199315
(Low Average) (Low Runs) 0.653333 0.951456 1.199315
(Low Strike Rate) (Low Average) 0.506667 0.844444 1.229773
(Low Average) (Low Strike Rate) 0.506667 0.737864 1.229773
(Medium Strike Rate) (Low Average) 0.106667 0.470588 0.685323
(Low Runs) (Low Strike Rate) 0.560000 0.705882 1.176471
(Low Strike Rate) (Low Runs) 0.560000 0.933333 1.176471
(Medium Average) (Low Runs) 0.100000 0.405405 0.511015
(Medium Strike Rate) (Low Runs) 0.153333 0.676471 0.852694

Strong Association Rules:
antecedents      consequents support confidence lift
(Fifty, Medium Average) (Medium Runs) 0.113333 0.607143 4.139610
(Fifty, Medium Runs) (Fifty, Medium Average) 0.113333 0.772727 4.139610
(Fifty, Medium Runs) (Medium Average) 0.113333 0.850000 3.445946
(Many Fours) (Fifty) 0.126667 1.000000 3.333333
(Medium Average, Medium Runs) (Fifty) 0.113333 1.000000 3.333333
(Medium Runs) (Medium Average) 0.113333 0.772727 3.132678
(Medium Runs) (Fifty) 0.133333 0.909091 3.030303
(Medium Average) (Fifty) 0.186667 0.756757 2.522523
(Fifty) (Medium Average) 0.186667 0.622222 2.522523
(High Strike Rate) (Fifty) 0.106667 0.615385 2.051282
(Medium Strike Rate) (Fifty) 0.113333 0.500000 1.666667
(Low Strike Rate, Low Runs) (Low Average) 0.500000 0.892857 1.300277
(Low Average) (Low Strike Rate, Low Runs) 0.500000 0.728155 1.300277
(Low Strike Rate) (Low Runs, Low Average) 0.500000 0.833333 1.275510
(Low Runs, Low Average) (Low Strike Rate) 0.500000 0.765306 1.275510
(Low Strike Rate, Low Average) (Low Runs) 0.500000 0.986842 1.243919
(Low Strike Rate, Low Average) (Low Strike Rate, Low Average) 0.500000 0.630252 1.243919

(Low Runs) (Low Average) 0.653333 0.823529 1.199315
(Low Average) (Low Runs) 0.653333 0.951456 1.199315
(Low Strike Rate) (Low Average) 0.506667 0.844444 1.229773
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(Medium Average) (Low Runs) 0.100000 0.405405 0.511015
(Medium Strike Rate) (Low Runs) 0.153333 0.676471 0.852694

Strong Association Rules:
antecedents      consequents support confidence lift
(Fifty, Medium Average) (Medium Runs) 0.113333 0.607143 4.139610
(Fifty, Medium Runs) (Fifty, Medium Average) 0.113333 0.772727 4.139610
(Fifty, Medium Runs) (Medium Average) 0.113333 0.850000 3.445946
(Many Fours) (Fifty) 0.126667 1.000000 3.333333
(Medium Average, Medium Runs) (Fifty) 0.113333 1.000000 3.333333
(Medium Runs) (Medium Average) 0.113333 0.772727 3.132678
(Medium Runs) (Fifty) 0.133333 0.909091 3.030303
(Medium Average) (Fifty) 0.186667 0.756757 2.522523
(Fifty) (Medium Average) 0.186667 0.622222 2.522523
(High Strike Rate) (Fifty) 0.106667 0.615385 2.051282
(Medium Strike Rate) (Fifty) 0.113333 0.500000 1.666667
(Low Strike Rate, Low Runs) (Low Average) 0.500000 0.892857 1.300277
(Low Average) (Low Strike Rate, Low Runs) 0.500000 0.728155 1.300277
(Low Strike Rate) (Low Runs, Low Average) 0.500000 0.833333 1.275510
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(Low Strike Rate, Low Average) (Low Strike Rate, Low Average) 0.500000 0.630252 1.243919
(Low Strike Rate) (Low Strike Rate, Low Average) 0.506667 0.844444 1.229773
(Low Average) (Low Strike Rate) 0.506667 0.737864 1.229773
(Low Runs) (Low Average) 0.653333 0.823529 1.199315

ASSOCIATION RULE MINING SUMMARY
Total transactions: 150
Number of frequent itemsets: 24
Number of association rules: 34
Number of strong rules: 23
```

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Figure 1

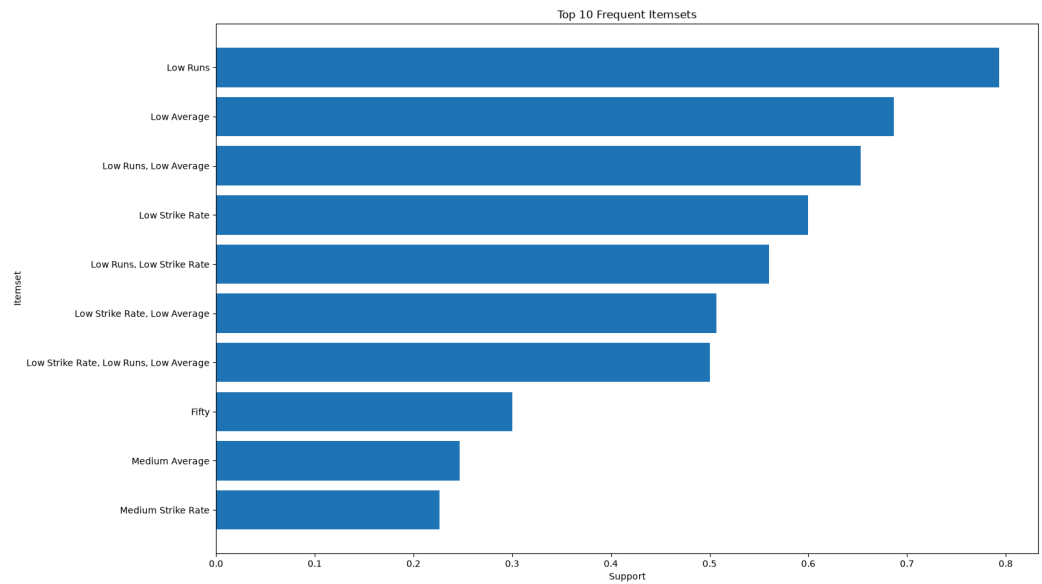
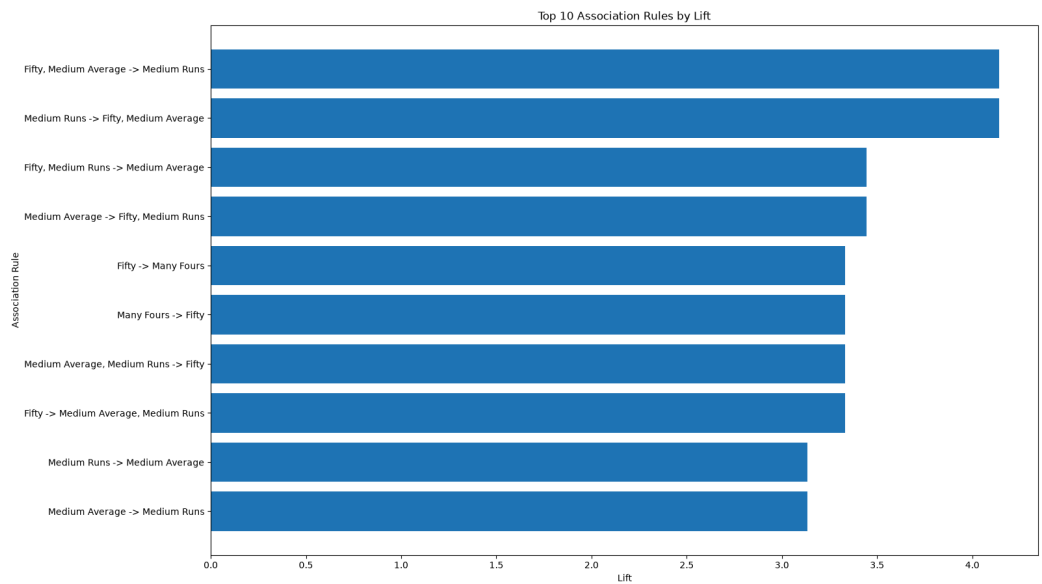


Figure 2



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